Lecture 3

Processes & Scheduling

How an operating system decides what runs next

Prof. A. Example · Winter term 2026

No notes

What is a process?

- A process is a program in execution.
  - It has its own address space: code, data, heap and stack.
- The OS keeps a Process Control Block (PCB) for each one.
  - PCB: process ID, state, program counter, registers, open files.
- Many processes share one CPU by taking turns.
- Switching between them is called a context switch.

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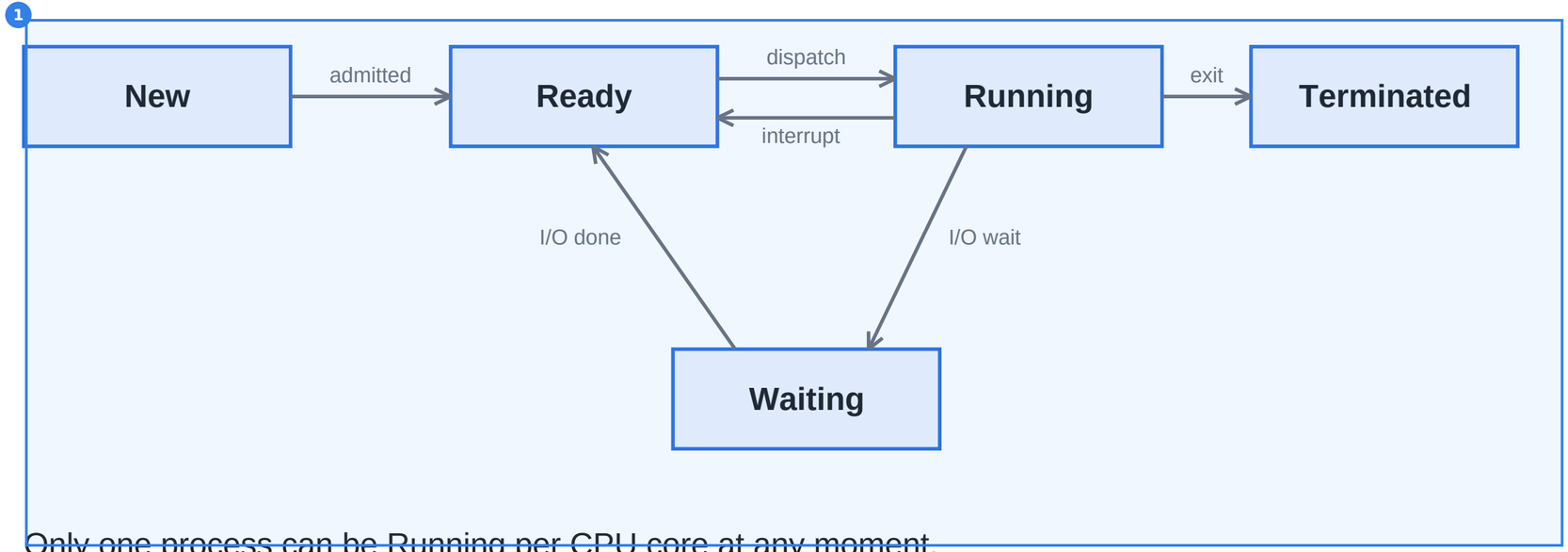
Prof said: context switches are pure overhead, no useful work happens. Likely exam question!

1 “Process Control Block (PCB)”

PCB = the OS's "ID card" for a process. Saved on every context switch.

The five process states



Only one process can be Running per CPU core at any moment.

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1 “New Ready Running Terminated Waiting admitted dispatch interrupt exit I/O wait I/O done”

Memorise all 5 states + the 6 arrows. Waiting → Ready, never straight to Running.

Processes vs. threads

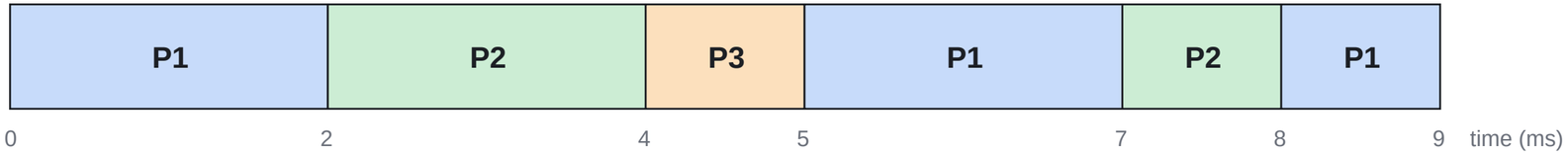
	Process	Thread
Memory	Own address space	Shared with its process
Creation cost	High (fork, new PCB)	Low
Communication	IPC: pipes, sockets	Shared variables
Crash impact	Isolated	Can take down all threads

No notes

CPU scheduling algorithms

- FCFS: first come, first served. Simple, but long jobs block.
- SJF: shortest job first.[1] Optimal average waiting time.
- Round Robin: each process gets a fixed time quantum.

Round Robin, quantum = 2



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1. “Optimal average waiting time.”
   Only optimal if burst times are known in advance (they usually aren't).

Summary

- A process = program + execution state (stored in the PCB).
- States: new, ready, running, waiting, terminated.
- Threads share memory, which makes them cheap but risky.
- The scheduler trades off fairness, throughput and response time.

Exam tip: be able to draw the state diagram and compute
average waiting time for FCFS, SJF and Round Robin.

No notes

Lecture notes

Summary
- Process = program + state (PCB)
- 5 states, know the transitions
- FCFS / SJF / RR: practise waiting-time calculations

Questions for the tutorial
- What happens to the PCB when a process terminates?